

Track 41: Network & System Administration

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Departments: IT · Cyber Security · CSE (elective) | MR ROKESH TECHNOLOGY

Skill path: Networking → TCP/IP → Linux/Windows Server → Active Directory → Virtualization → Cloud Admin → Security Hardening

Related: [Track 06 — Cybersecurity](#) | [Track 07 — DevOps](#)

Track Overview

Who this is for: IT students targeting network administrator, system administrator, and IT infrastructure roles.

Prerequisites: Basic computer literacy; Linux introduced from Week 1.

Tools: Linux (Ubuntu/CentOS), Windows Server, VirtualBox/VMware, Active Directory, DNS/DHCP, Wireshark, AWS/Azure admin consoles.

Certification targets: CompTIA Network+, Linux+ (optional, 6-month)

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	IT infra awareness	Configured lab network
1 Month	80–100	Junior sysadmin basics	Linux + Windows server setup
3 Months	250–300	Internship-ready	Enterprise lab environment
6 Months	500–600	Sysadmin/network admin ready	Admin portfolio + cert prep

2-Week Crash Course

Days	Topics	Deliverable
D1–2	OSI/TCP-IP, IP addressing, subnetting	Subnetting worksheet
D3–4	Linux admin: users, permissions, services	Configured Linux server
D5–6	Windows Server, Active Directory basics	AD domain with 3 users
D7–8	DNS, DHCP, file sharing, SSH	Working lab network
D9–10	Wireshark, troubleshooting, capstone	Network documentation + lab demo

1-Month Foundation

Week	Topics	Project
W1	Linux deep dive, shell scripting, cron	Automated admin scripts
W2	Windows Server roles, GPO intro	Managed desktop environment
W3	Virtualization, backup/restore	VM infrastructure
W4	Network security hardening, firewall	Secure lab environment

3-Month Job Ready

Month	Topics	Projects
M1	Advanced networking, routing, VLANs	Multi-VLAN lab
M2	Cloud admin (AWS/Azure IAM, EC2, VMs)	Hybrid cloud lab
M3	Monitoring (Nagios/Zabbix intro), capstone	Enterprise IT infrastructure capstone

6-Month Professional

Add: VPN setup, disaster recovery, automation (Ansible for admin), Network+/Security+ prep, incident response basics.

Assessment Rubric

Weekly Quiz 20% | Lab configuration 40% | Project 30% | Portfolio 10%

Appendix: Mentor Notes

All labs in isolated VMs. Foundation for [Track 06](#) and [Track 44](#).